Computing for Bioinformatics Lab Manual

Lab: Biological Network Analysis with Cytoscape

In class we learnt that genes and proteins do not work alone. They interact with each other in large networks, and some proteins sit at the centre of these networks, connected to many others. Today you will build and analyse real networks using **Cytoscape**

By the end of this lab, you will have built and analysed both networks, found their hub proteins, and built a network of your own for a human, animal or plant question.

Biological Network Terms?

Network term	Protein network
Node	A protein (or gene)
Edge	An interaction between two proteins
Degree	Number of proteins a protein interacts with
Hub	A highly connected protein; often a key controller of a response
Betweenness centrality	A protein that connects different parts of the network (a bridge)
Cluster (module)	A group of proteins working together on one function

Today the interaction data will come from **STRING** (string-db.org), a database that collects known and predicted protein interactions for thousands of organisms, including plants, each with a confidence score.

Key Concepts to Note

1. An edge in STRING is evidence, not a fact

STRING combines evidence from experiments, curated databases, text mining of publications, co-expression and other sources. Each edge has a **confidence score** between 0 and 1. A score of 0.9 means strong evidence; 0.4 (the default) is medium confidence.

Note:

A protein with few edges is not necessarily unimportant. It may simply be less studied. Well-studied proteins collect more edges, so always think about this bias when you call something a hub. This matters even more for less-studied organisms such as cattle or sorghum, compared with humans or Arabidopsis.

2. Hubs and bottlenecks

A **hub** has a high degree (many direct partners). A **bottleneck** has high betweenness centrality (many shortest paths pass through it). Proteins that are both are often the master switches of a response, which makes them prime targets for crop improvement.

3. Always choose the right species

Many gene names are shared across species. Before every query, check the species: **Homo sapiens** for Example 1 and **Arabidopsis thaliana** for Example 2. Otherwise STRING may return a protein from the wrong organism with a similar name.

Setting Up for Today's Lab

Cytoscape is a desktop application. It runs directly on Windows or macOS, **not** inside WSL Ubuntu.

1. Download the latest version of Cytoscape from cytoscape.org/download.html and install it. The installer includes Java, so no separate installation is needed.
2. Open Cytoscape. The first launch can take a minute.
3. Install the stringApp: go to **Apps → App Store → Show App Store**, search for **stringApp**, and click **Install**.
4. Check that you are connected to the internet: STRING data is downloaded live during the lab.

Example 1: A Human Cancer Network (TP53)

Goal: build the protein interaction network around **TP53** and learn how to read, arrange and style a network in Cytoscape.

TP53 makes a protein called p53, often called the "guardian of the genome". When a cell's DNA is damaged, p53 acts like a security guard at a gate: it stops the cell from dividing until the damage is repaired, or orders the cell to destroy itself if the damage is too great. When TP53 is mutated, the guard is gone, and damaged cells can keep dividing. This is why TP53 is mutated in about half of all human cancers, including breast and cervical cancer, two of the most common cancers in Kenya.

Import the network from STRING

Step 1: In the **Network** panel on the left (Control Panel), find the search bar at the top. From its drop-down, choose **STRING protein query**. (You can also use **File → Import → Network from Public Databases**.)

Step 2: Type **TP53** in the search box.

Step 3: Click the options button beside the search box and set the species to **Homo sapiens**. Set the maximum number of additional interactors to **10**. Leave the confidence score at its default (0.4).

Step 4: Click the search (magnifying glass) button. If a dialog asks you to confirm the protein, choose **TP53 (human)** and click **Import**.

Cytoscape draws the network: TP53 plus its 10 strongest interaction partners. Each circle is a protein; each line is an interaction.

Take a screenshot of your network at this point.

Explore the network

Step 5: Click a node. Look at the **Node Table** at the bottom of the window: it shows the protein's name, full description and other details from STRING.

Step 6: Click an edge. In the **Edge Table**, look at the score columns. Which type of evidence supports this interaction most strongly?

Step 7: Try a different layout: go to **Layout → Prefuse Force Directed Layout**. Highly connected proteins are pulled towards the centre.

Note : Layouts only change how the network looks, not the data. Nodes in the middle of a force-directed layout tend to be the well-connected ones.

Calculate the network statistics

Step 8: Go to **Tools → Analyze Network**. If asked, analyse it as an **undirected** network (protein interactions in STRING have no direction).

Step 9: Look at the **Node Table**. New columns have been added, including **Degree** and **BetweennessCentrality**. Click the **Degree** header (twice) to sort from highest to lowest.

Take a screenshot of your Node Table sorted by Degree.

Questions to answer

1. How many nodes and edges does your network have? (Look at the network name in the Network panel.)
2. Which protein has the highest Degree? Is it TP53? Why would you expect this?
3. Pick three proteins connected to TP53. Using the description in the Node Table, write one sentence on what each one does.
4. Which edge has the highest combined score?

Example 2: A Plant Drought-Response Network

Our second story is about **drought**. When the rains fail in Makueni or Kitui, a maize plant cannot walk to find water. Instead, a hormone called **abscisic acid (ABA)** acts as an alarm: it tells the leaves to close their tiny pores (stomata) to save water and switches on hundreds of protective genes. The proteins that receive and pass on this alarm form a network, and crop scientists study it to breed drought-tolerant varieties.

We will use **Arabidopsis thaliana**, a small weed that is the "lab mouse" of plant biology. Its genes are the best studied of any plant, and most of what we learn from it applies to crops like maize, sorghum and beans.

Aim: build a larger network from ten genes involved in ABA drought signalling, from making the hormone, to sensing it, to switching on protective genes, then calculate network statistics and let the data show which proteins are the hubs.

Import a multi-gene network

Step 1: Open a new STRING protein query (as in Example 1), and change the species to **Arabidopsis thaliana**. This time, type in the ten gene names from the first column of the table below, one per line:

Gene	Role in the drought alarm
NCED3	Makes ABA (the alarm hormone)
PYR1	ABA receptor (hears the alarm)
ABI1	Brake that keeps the response off
ABI2	Brake (partner of ABI1)
HAB1	Brake (same family as ABI1)
OST1	Kinase that closes the stomata
ABF2	Transcription factor that switches on stress genes
ABI5	Transcription factor (seed and stress response)
DREB2A	Transcription factor for drought and heat

Gene	Role in the drought alarm
RD29A	Protective gene switched on by drought

Step 2: Check that the species is **Arabidopsis thaliana** and set the additional interactors to **10**. Click search. If STRING asks you to confirm any names, choose the Arabidopsis entries, then click **Import**.

Take a screenshot of your network at this point.

Calculate the network statistics

Step 3: Go to **Tools → Analyze Network**. If asked, analyse it as an **undirected** network (protein interactions in STRING have no direction).

Step 4: When it finishes, look at the **Node Table**. New columns have been added, including **Degree**, **BetweennessCentrality** and **ClosenessCentrality**.

Step 5: Click the **Degree** column header to sort it, then click again so the highest values are at the top. The proteins at the top are your hubs.

Note : Analyze Network must be run again every time you add or remove nodes, otherwise the numbers in the table are out of date.

Make the hubs visible

Step 6: Open the **Style** tab in the Control Panel. Find **Size**, click the arrow on its right, set **Column** to **Degree** and **Mapping Type** to **Continuous Mapping**. High-degree nodes now appear larger. (If size does not change, tick **Lock node width and height**.)

Step 7: Do the same for **Fill Color**: map it to **BetweennessCentrality** with a continuous mapping, so the main bridges stand out in a stronger colour.

Step 8: Apply **Layout → Prefuse Force Directed Layout** again.

Take a screenshot of your styled network showing the hubs.

What does this network do? (Functional enrichment)

Step 9: Click on an empty part of the canvas so no nodes are selected. Go to **Apps → STRING Enrichment → Retrieve functional enrichment** and keep the defaults.

Step 10: A **STRING Enrichment** tab opens in the table panel. It lists biological processes and pathways that are over-represented in your network, with an FDR value (the smaller, the more significant).

Take a screenshot of the top rows of the enrichment table.

Save your work

Step 11: Export the network picture: **File → Export → Network to Image**, and save it as a PNG.

Step 12: Save the whole session (network, styles and tables): **File → Save As**. This creates a .cys file you can reopen later.

Questions to answer

1. List the top 5 proteins by Degree and the top 5 by BetweennessCentrality. Are they the same proteins?
2. Are the "brakes" (ABI1, ABI2, HAB1) hubs in this network? Use their Degree to justify your answer.
3. Name two enriched biological processes. Do they make sense for a plant responding to drought?
4. If you were a plant breeder at KALRO trying to develop a drought-tolerant maize variety, which hub protein would you study first, and why?

Exercise : Your Own Network (You Design This)

Now apply the same steps to a new question, on your own. Choose **one** option from the table below: a human, animal or plant question. You may also use a gene from your group project (agree it with your lecturer).

3.1 Choose your network

Question	Species in STRING	Suggested starting gene(s)
Breast cancer	Homo sapiens	BRCA1, BRCA2, TP53
Sickle cell disease	Homo sapiens	HBB
Type 2 diabetes	Homo sapiens	INS, INSR, TCF7L2
Tuberculosis (host immune response)	Homo sapiens	TLR2, VDR, IFNG
Obesity and appetite control	Mus musculus (mouse)	Lep, Lepr, Mc4r
Milk production in dairy cattle	Bos taurus (cattle)	CSN2, LALBA, STAT5A
Salt tolerance	Arabidopsis thaliana	SOS1, SOS2, SOS3
Flowering time	Arabidopsis thaliana	FLC, CO, FT, SOC1
Disease resistance	Arabidopsis thaliana	NPR1, FLS2, EDS1

Problem statement

For your chosen option:

1. Build a protein interaction network in Cytoscape using STRING, with the correct species, starting from your chosen genes with up to 20 additional interactors.
2. Run Analyze Network and identify the top 5 hub proteins by Degree and the top 5 by BetweennessCentrality.
3. Style the network so that node size shows Degree and node colour shows BetweennessCentrality.
4. Run functional enrichment and report the three most significant biological processes or pathways.
5. Export a clear image of your final network.

Requirements checklist

- The species must match your chosen option (check it before every query).
- Analyze Network must be run on the final version of your network.
- Fill in a results table like the one below for your top 5 hubs.

Rank	Protein	Degree	Betweenness
1			
2			
3			
4			
5			

- Write one paragraph explaining whether your top hub makes biological sense for your chosen question, and one limitation of your network

Submission

Submit a Word document containing:

- The required screenshots from Examples 1 and 2, and your answers to their questions.
- For the Exercise:
    - a screenshot of your final styled network;
    - a screenshot of your Node Table sorted by Degree;
    - a screenshot of your enrichment results;
    - your completed hub table and paragraph.
- A brief reflection on what you have learnt from this lab and what was challenging.
- Submit before the deadline.

Enjoy Your Day!